

Individual Demand, Income and Substitution Effects & Market Demand

ECON 101: Introduction to
Microeconomics

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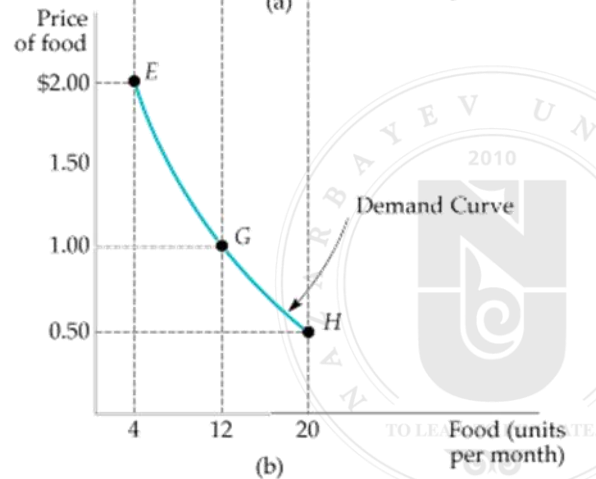
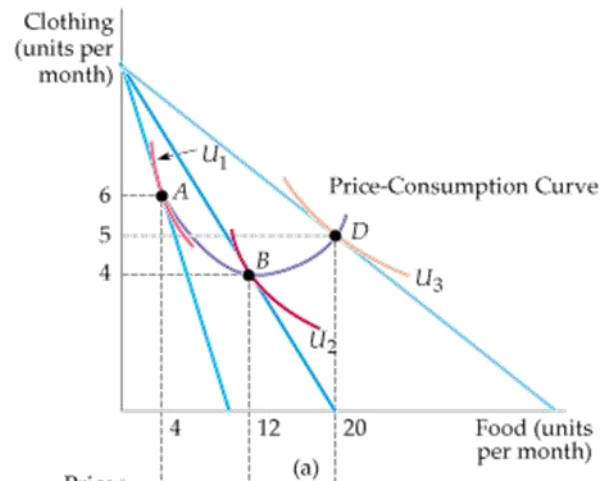
Individual Demand

EFFECT OF PRICE CHANGES

A reduction in the price of food, with income and the price of clothing fixed, causes the consumer to choose a different market basket.

In panel **(a)**, the baskets that maximize utility for various prices of food (point A, \$2; B, \$1; D, \$0.50) trace out the price-consumption curve.

Part **(b)** gives the demand curve, which relates the price of food to the quantity demanded. (Points E, G, and H correspond to points A, B, and D, respectively).



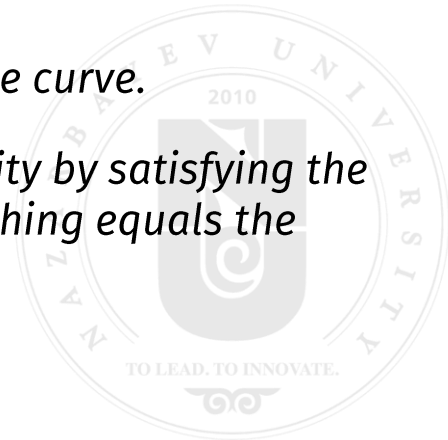
The individual demand curve

price-consumption curve Curve tracing the utility-maximizing combinations of two goods as the price of one changes.

individual demand curve Curve relating the quantity of a good that a single consumer will buy to its price.

The individual demand curve has two important properties:

1. *The level of utility that can be attained changes as we move along the curve.*
2. *At every point on the demand curve, the consumer is maximizing utility by satisfying the condition that the marginal rate of substitution (MRS) of food for clothing equals the ratio of the prices of food and clothing.*



Income changes

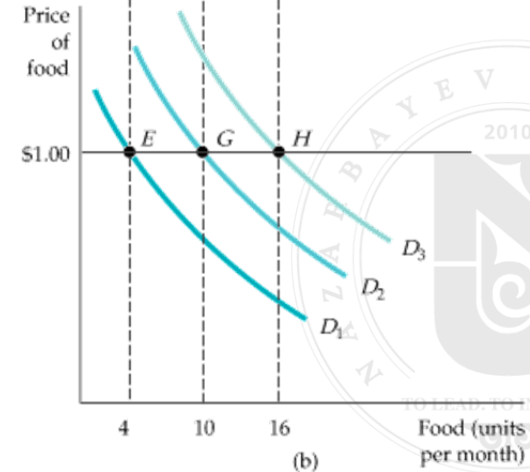
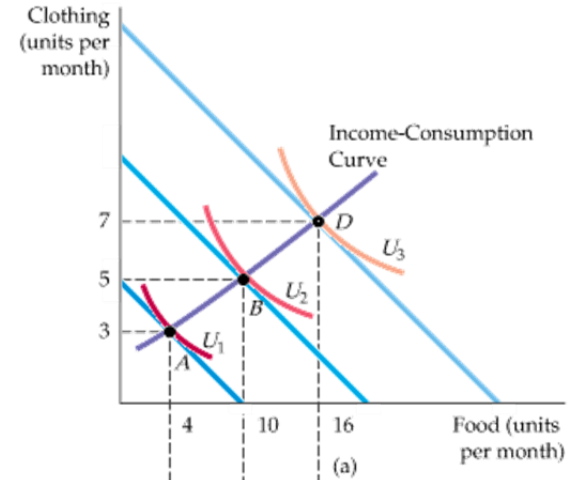
EFFECT OF INCOME CHANGES

An increase in income, with the prices of all goods fixed, causes consumers to alter their choice of market baskets.

In part (a), the baskets that maximize consumer satisfaction for various incomes (point A, \$10; B, \$20; D, \$30) trace out the income-consumption curve.

The shift to the right of the demand curve in response to the increases in income is shown in part (b). (Points E, G, and H correspond to points A, B, and D, respectively.)

income-consumption curve Curve tracing the utility-maximizing combinations of two goods as a consumer's income changes.

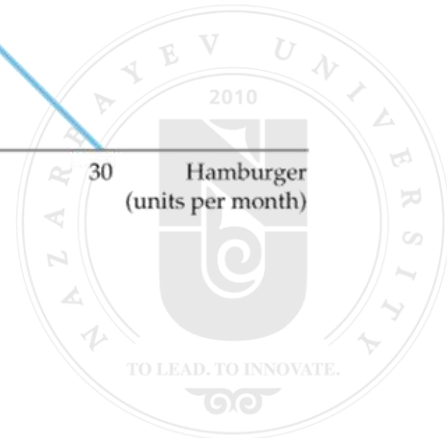
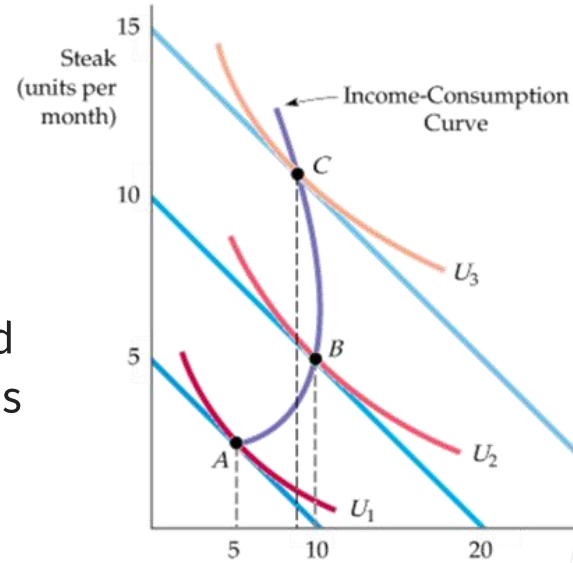


Normal vs Inferior Goods

AN INFERIOR GOOD

An increase in a person's income can lead to less consumption of one of the two goods being purchased.

Here, hamburger, though a normal good between *A* and *B*, becomes an inferior good when the income-consumption curve bends backward between *B* and *C*.



Engel Curves

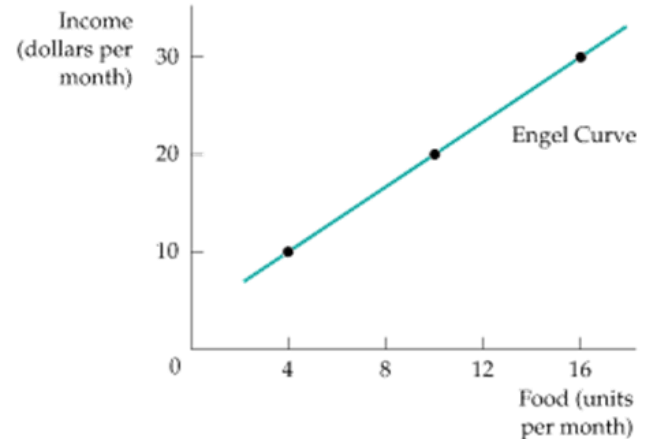
ENGLE CURVES

Engel curves relate the quantity of a good consumed to income.

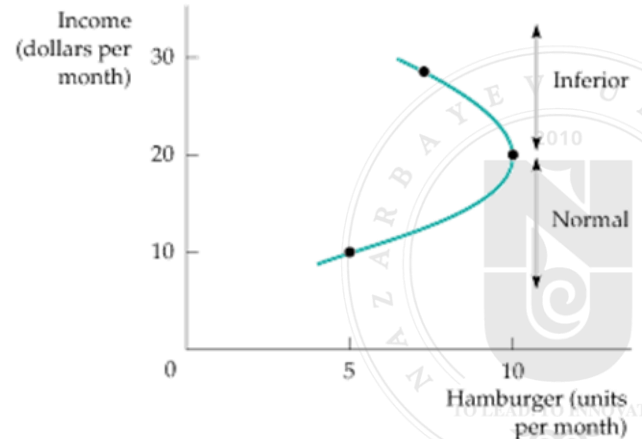
In **(a)**, food is a normal good and the Engel curve is upward sloping.

In **(b)**, however, hamburger is a normal good for income less than \$20 per month and an inferior good for income greater than \$20 per month.

Engel curve Curve relating the quantity of a good consumed to income.



(a)



(b)

Example: Consumer expenditures in the US

- We can derive Engel curves for groups of consumers. This information is particularly useful if we want to see how consumer spending varies among different income groups.

EXPENDITURES (\$) ON:	LESS THAN \$10,000	10,000– 19,999	20,000– 29,999	30,000– 39,999	40,000– 49,999	50,000– 69,999	70,000 AND ABOVE
Entertainment	1,038	1,165	1,407	1,969	2,131	2,548	4,655
Owned Dwelling	1,770	2,134	2,795	3,581	4,198	5,556	11,606
Rented Dwelling	3,919	3,657	4,054	3,878	4,273	3,812	3,072
Health Care	1,434	2,319	3,124	3,539	3,709	4,702	6,417
Food	3,466	3,706	4,432	5,194	5,936	6,486	10,116
Clothing	798	766	960	1,321	1,518	1,602	2,928

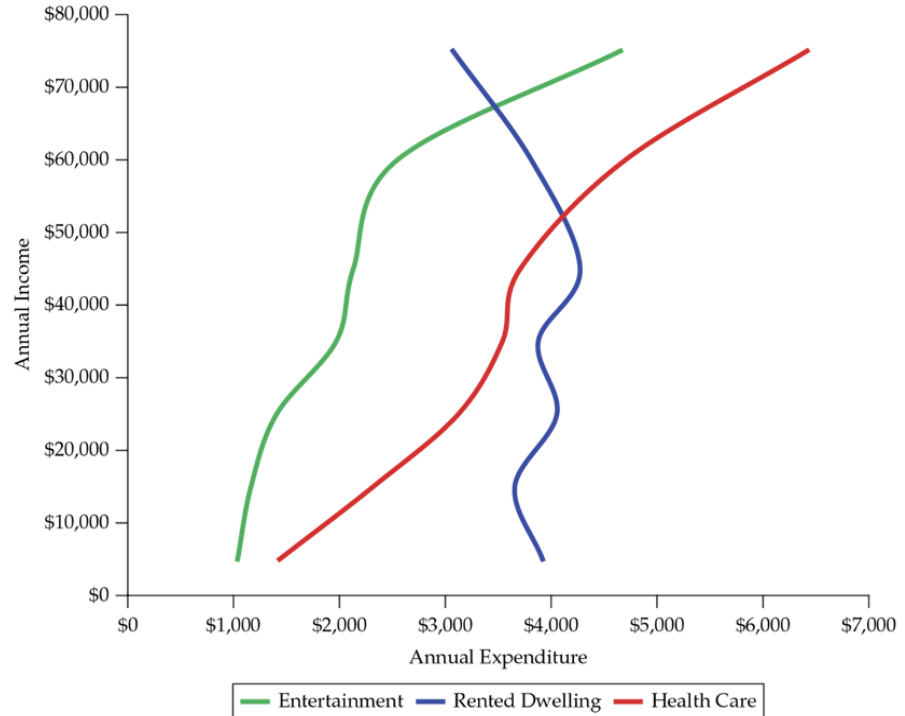
Example: Consumer expenditures in the US

ENGEL CURVES FOR U.S. CONSUMERS

Average per-household expenditures on rented dwellings, health care, and entertainment are plotted as functions of annual income.

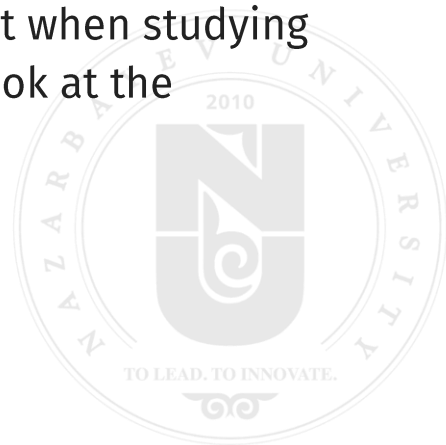
Health care and entertainment are normal goods, as expenditures increase with income.

Rental housing, however, is an inferior good for incomes above \$40,000.



Substitutes and Complements

- Two goods are *substitutes* if an increase in the price of one leads to an increase in the quantity demanded of the other.
- Two goods are *complements* if an increase in the price of one good leads to a decrease in the quantity demanded of the other.
- Two goods are *independent* if a change in the price of one good has no effect on the quantity demanded of the other.
- The fact that goods can be complements or substitutes suggests that when studying the effects of price changes in one market, it may be important to look at the consequences in related markets.



Income and Substitution Effects

A fall in the price of a good has two effects:

1. *Consumers will tend to buy more of the good that has become cheaper and less of those goods that are now relatively more expensive.* This response to a change in the relative prices of goods is called the *substitution effect*.
2. *Because one of the goods is now cheaper, consumers enjoy an increase in real purchasing power.* The change in demand resulting from this change in real purchasing power is called the *income effect*.



Income and Substitution Effects

Substitution Effect

substitution effect Change in consumption of a good associated with a change in its price, with the level of utility held constant.

Income Effect

income effect Change in consumption of a good resulting from an increase in purchasing power, with relative prices held constant.



Income and Substitution Effects

INCOME AND SUBSTITUTION EFFECTS: NORMAL GOOD

A decrease in the price of food has both an income effect and a substitution effect.

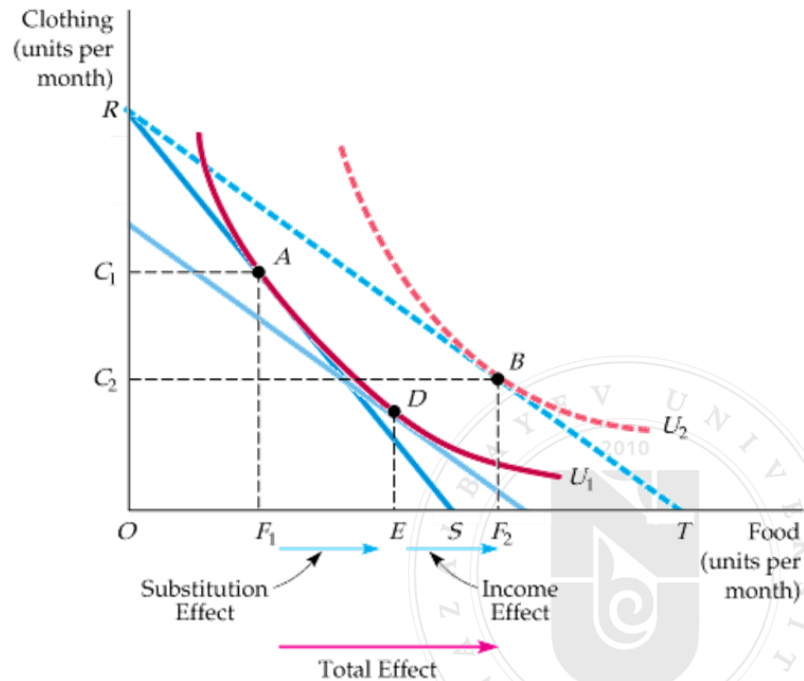
The consumer is initially at A , on budget line RS .

When the price of food falls, consumption increases by F_1F_2 as the consumer moves to B .

The substitution effect F_1E (associated with a move from A to D) changes the relative prices of food and clothing but keeps real income (satisfaction) constant.

The income effect EF_2 (associated with a move from D to B) keeps relative prices constant but increases purchasing power.

Food is a normal good because the income effect EF_2 is positive.



Income and Substitution Effects

INCOME AND SUBSTITUTION EFFECTS: INFERIOR GOOD

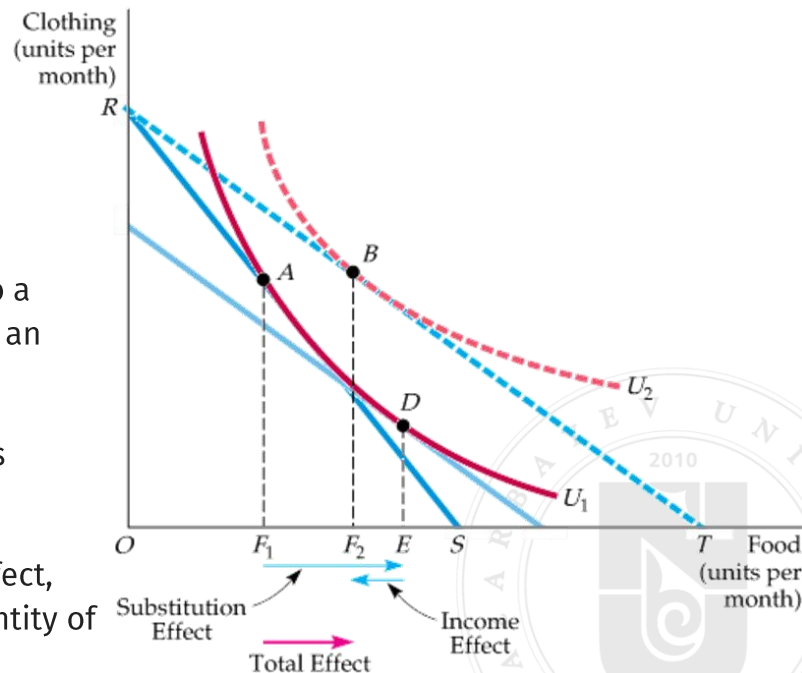
The consumer is initially at A on budget line RS .

With a decrease in the price of food, the consumer moves to B .

The resulting change in food purchased can be broken down into a substitution effect, F_1E (associated with a move from A to D), and an income effect, EF_2 (associated with a move from D to B).

In this case, food is an inferior good because the income effect is negative.

However, because the substitution effect exceeds the income effect, the decrease in the price of food leads to an increase in the quantity of food demanded.



Income and Substitution Effects

A Special Case: The Giffen Good

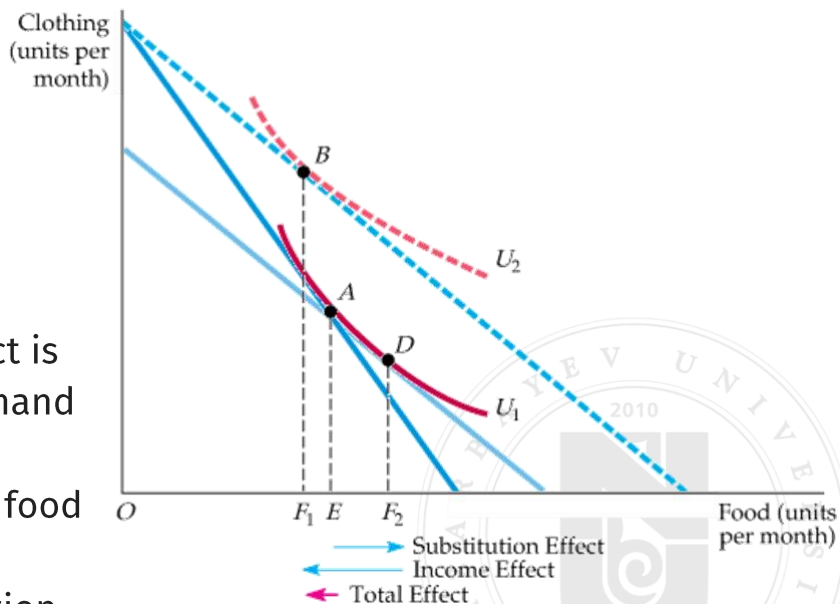
Giffen good Good whose demand curve slopes upward because the (negative) income effect is larger than the substitution effect.

UPWARD-SLOPING DEMAND CURVE: THE GIFFEN GOOD

When food is an inferior good, and when the income effect is large enough to dominate the substitution effect, the demand curve will be upward-sloping.

The consumer is initially at point A, but, after the price of food falls, moves to B and consumes less food.

Because the income effect F_2F_1 is larger than the substitution effect EF_2 , the decrease in the price of food leads to a lower quantity of food demanded.



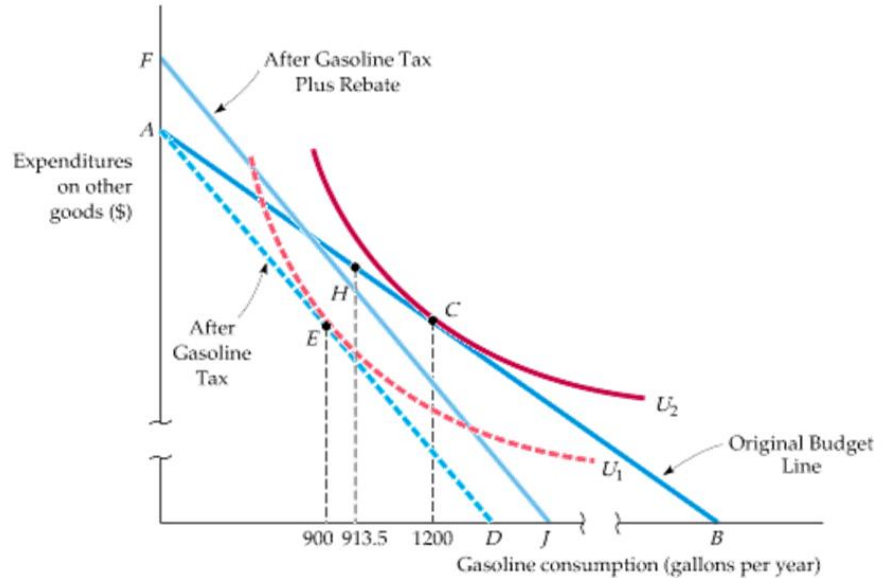
Example: The effects of a Gasoline Tax

EFFECT OF A GASOLINE TAX WITH A REBATE

A gasoline tax is imposed when the consumer is initially buying 1200 gallons of gasoline at point C. After the tax takes effect, the budget line shifts from AB to AD and the consumer maximizes his preferences by choosing E, with a gasoline consumption of 900 gallons.

However, when the proceeds of the tax are rebated to the consumer, his consumption increases somewhat, to 913.5 gallons at H.

Despite the rebate program, the consumer's gasoline consumption has fallen, as has his level of satisfaction.



Market Demand

market demand curve Curve relating the quantity of a good that all consumers in a market will buy to its price.

From Individual to Market Demand

(1) PRICE (\$)	(2) INDIVIDUAL A (UNITS)	(3) INDIVIDUAL B (UNITS)	(4) INDIVIDUAL C (UNITS)	(5) MARKET UNITS
1	6	10	16	32
2	4	8	13	25
3	2	6	10	18
4	0	4	7	11
5	0	2	4	6

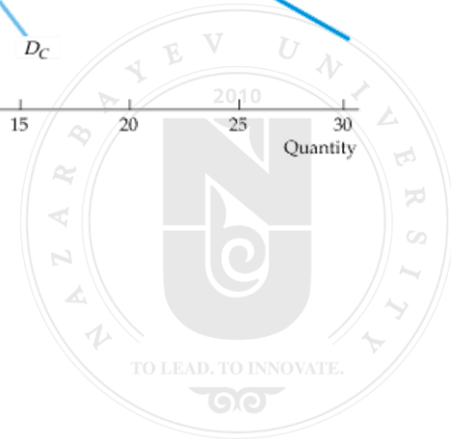
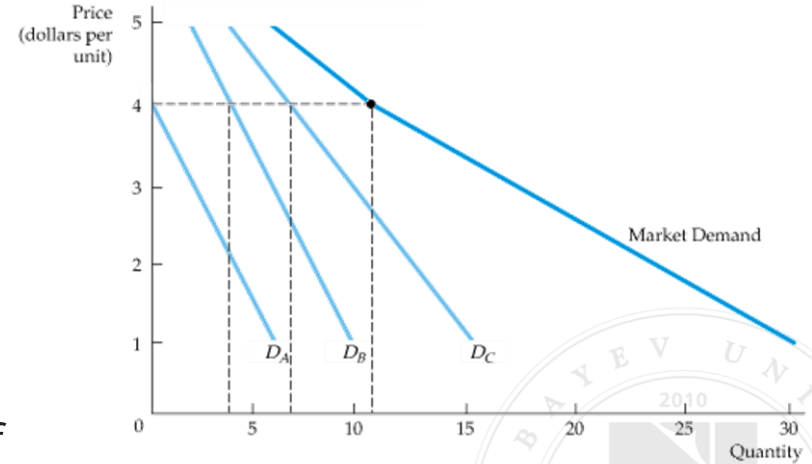
Market Demand

SUMMING TO OBTAIN A MARKET DEMAND CURVE

The market demand curve is obtained by summing our three consumers' demand curves D_A , D_B , and D_C .

At each price, the quantity of coffee demanded by the market is the sum of the quantities demanded by each consumer.

At a price of \$4, for example, the quantity demanded by the market (11 units) is the sum of the quantity demanded by A (no units), B (4 units), and C (7 units).

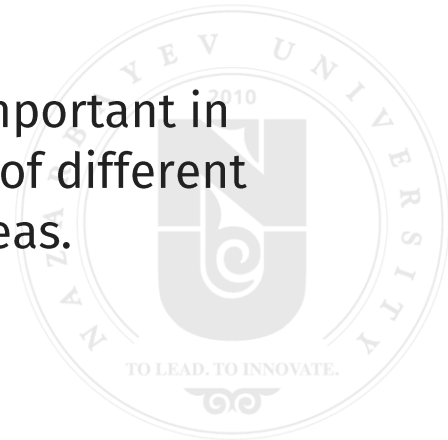


Market Demand

Two points should be noted:

1. *The market demand curve will shift to the right as more consumers enter the market.*
2. *Factors that influence the demands of many consumers will also affect market demand.*

The aggregation of individual demands into market becomes important in practice when market demands are built up from the demands of different demographic groups or from consumers located in different areas.



Example: Aggregate Demand for Wheat

Domestic demand for wheat is given by the equation

$$Q_{DD} = 1430 - 55P$$

where Q_{DD} is the number of bushels (in millions) demanded domestically, and P is the price in dollars per bushel.

Export demand is given by

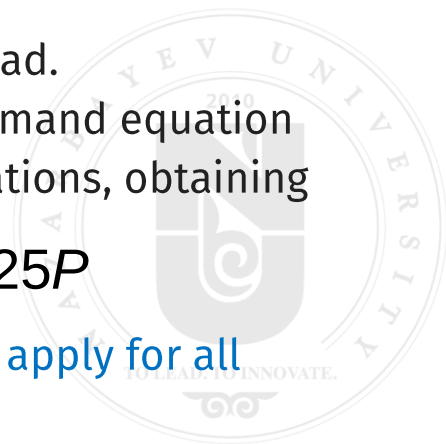
$$Q_{DE} = 1470 - 70P$$

where Q_{DE} is the number of bushels (in millions) demanded from abroad.

To obtain the world demand for wheat, we set the left side of each demand equation equal to the quantity of wheat. We then add the right side of the equations, obtaining

$$Q_{DD} + Q_{DE} = (1430 - 55P) + (1470 - 70P) = 2900 - 125P$$

However, we must be careful, because the expression above does not apply for all prices! Aggregate demand can be kinked!

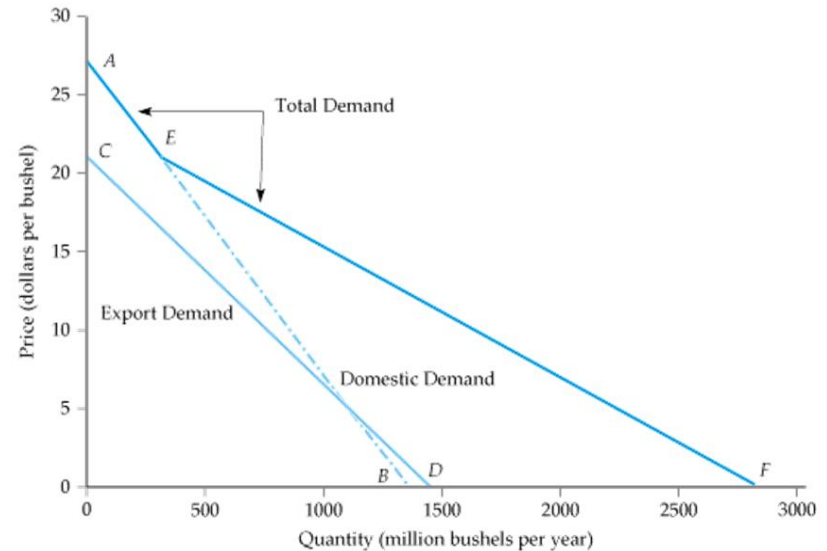


Example: Aggregate Demand for Wheat

THE AGGREGATE DEMAND FOR WHEAT

The total world demand for wheat is the horizontal sum of the domestic demand AB and the export demand CD .

Even though each individual demand curve is linear, the market demand curve is kinked, reflecting the fact that there is no export demand when the price of wheat is greater than about \$21 per bushel.



Example: The demand for housing, aggregating across age/income groups.

- There are significant differences in price and income elasticities of housing demand among subgroups of the population.

GROUP	PRICE ELASTICITY	INCOME ELASTICITY
Single individuals	– 0.10	0.21
Married, head of household age less than 30, 1 child	– 0.25	0.06
Married, head age 30–39, 2 or more children	– 0.15	0.12
Married, head age 50 or older, 1 child	– 0.08	0.19

- In recent years, the demand for housing has been partly driven by speculative demand. Speculative demand is driven not by the direct benefits one obtains from owning a home but instead by an expectation that the price will increase.



End of this Lecture.

Note: This lecture is based on parts of Chapter 4 from Pindyck & Rubinfeld.
However, the discussion on individual demand and income and substitution effects is carried out at a greater depth than in the book.